



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2022 Year 12 Course Assessment Task 4 (Trial Examination)

Tuesday August 16, 2022

General instructions

- Working time – 2 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 9)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MAX.1 – Mr Lam

☐ 12MXX.1 – Ms Ham

☐ 12MAX.2 – Ms C. Kim

☐ 12MAX.3 – Ms Lee

☐ 12MXX.2 – Mr Sekaran

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{18}$	$\overline{14}$	$\overline{13}$	$\overline{15}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 9).

Questions

Marks

1. Which of the following is the next step in progressing to the correct solution to the following inequality? 1

$$\frac{x^2 - 4}{x} \geq 0$$

(A) $x^2 - 4 \geq 0$

(C) $x^2 - 4 \leq 0$

(B) $x(x^2 - 4) \geq 0$

(D) $x(x^2 - 4) \leq 0$

2. Given $\tan \theta = \frac{1}{3}$, what is the exact value of 1

$$\tan \left(\theta + \frac{\pi}{3} \right)$$

(A) $\frac{\sqrt{3} + 3}{3\sqrt{3} - 1}$

(C) $\frac{1 + 3\sqrt{3}}{3 - \sqrt{3}}$

(B) $\frac{\sqrt{3} - 3}{3\sqrt{3} + 1}$

(D) $\frac{1 - 3\sqrt{3}}{3 + \sqrt{3}}$

3. Which Cartesian equation is equivalent to the parametric equations $x = 5 \cos \theta$ and $y = 6 \sin \theta$? 1

(A) $\frac{x}{5} + \frac{y}{6} = 1$

(C) $\frac{x^2}{25} + \frac{y^2}{36} = 1$

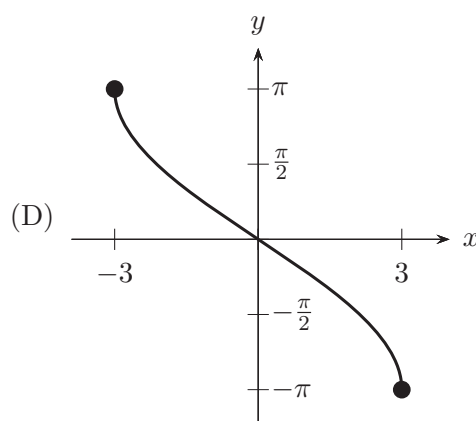
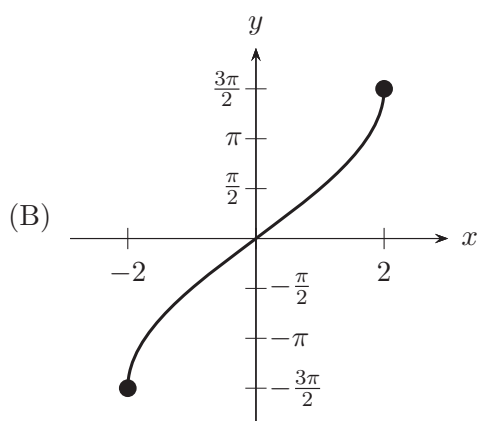
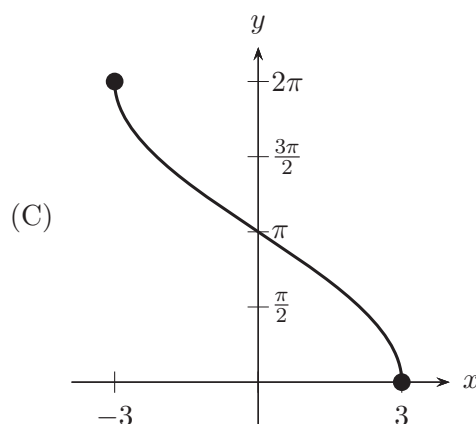
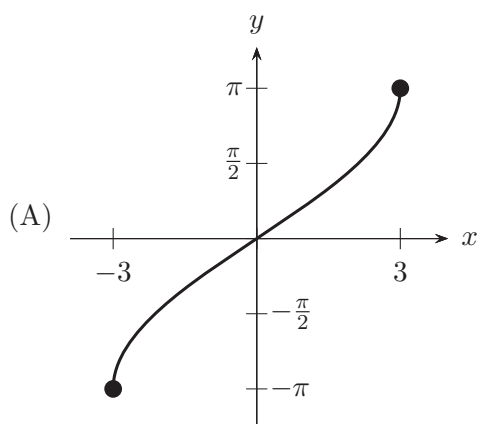
(B) $\frac{x}{25} + \frac{y}{36} = 1$

(D) $x^2 + y^2 = 61$

Examination continues overleaf...

4. Which graph best represents $y = 2 \sin^{-1} \frac{x}{3}$?

1



5. The equation $x^3 - 2x^2 + 4x - 3 = 0$ has roots $x = \alpha$, $x = \beta$ and $x = \gamma$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

1

(A) $\frac{4}{3}$

(C) $\frac{3}{4}$

(B) $-\frac{4}{3}$

(D) $-\frac{3}{4}$

6. Let $\underline{a}$ and $\underline{b}$ be two non-zero vectors which are neither parallel nor perpendicular such that $\underline{c} = \text{proj}_{\underline{b}} \underline{a}$. Which statement is always TRUE?

1

(A) $(\underline{c} + \underline{a}) \cdot \underline{b} = 0$

(C) $(\underline{a} + \underline{b}) \cdot \underline{c} = 0$

(B) $(\underline{a} - \underline{b}) \cdot \underline{c} = 0$

(D) $(\underline{c} - \underline{a}) \cdot \underline{b} = 0$

Examination continues overleaf...

7. Which of the following is the derivative of $f(x) = (\sin x)(\sin^{-1} x)$? 1

(A) $f'(x) = -\cos x \sin^{-1} x + \frac{\sin x}{\sqrt{1-x^2}}$

(B) $f'(x) = \cos x \sin^{-1} x + \frac{\sin x}{\sqrt{1-x^2}}$

(C) $f'(x) = -\sin x \sin^{-1} x + \frac{\cos x}{\sqrt{1-x^2}}$

(D) $f'(x) = \cos x \sin^{-1} x - \frac{\sin x}{\sqrt{1-x^2}}$

8. What is the primitive of $\frac{2f'(x)}{1+4(f(x)^2)}$? 1

(A) $\tan^{-1}(f(2x))$

(C) $2 \ln 2(f(x))$

(B) $2 \tan^{-1}(f(x))$

(D) $\tan^{-1}(2f(x))$

9. The expression $3 \cos x + 2 \sin x$ is written in the form $R \cos(x - \theta)$, where $R > 0$ and $0 \leq \theta \leq \frac{\pi}{2}$. 1

What is the value of $\tan \theta$?

(A) $-\frac{3}{2}$

(C) $\frac{2}{3}$

(B) $-\frac{2}{3}$

(D) $\frac{3}{2}$

10. What is the area enclosed between the curve $y = x^3 - 1$, the y -axis and the lines $y = 1$ and $y = 2$? 1

(A) $\int_1^2 (x^3 - 1) dy$

(C) $\int_1^2 (\sqrt{3}y + 1) dy$

(B) $\int_1^2 \sqrt[3]{y+1} dy$

(D) $\int_1^2 (y+1) dy$

Examination continues overleaf...

Section II

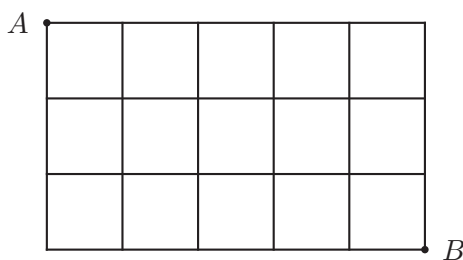
60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (18 Marks)	Commence a NEW booklet.	Marks
(a) i. Show that	$\frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} = \frac{1}{(n+2)!}$	1
ii. Use mathematical induction to prove that, for all integers $n \geq 1$,	$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$	3
(b) Find the constant term in the expansion of $\left(3x - \frac{1}{x^2}\right)^6$.		3
You may leave your answers in the form $\binom{n}{k}a^b$ where $n, k, a, b \in \mathbb{Z}^+$.		
(c) A school has twenty prefects. Jun and Alex are two of those prefects. A committee of seven prefects is to be formed to organise a function.		
i. How many different committees can be formed?		1
ii. What is the probability that the committee contains Jun but does not contain Alex?		2
(d) A eleven-member Fund Raising Committee consists of five students, four teachers and two parents. The committee meets around a circular table.		
i. How many different arrangements of the eleven members around the table are possible if the students sit together as a group and so do the teachers, but no teacher sits next to a student?		3
ii. One of the students is the child of one of the parents present. Given that all arrangements in (i) are equally likely, what is the probability that the student sits next to their parent?		3
(e) The diagram shows a grid consisting of unit squares. Chris needs to travel from point A to point B but can only do so by moving right or down along the grid lines.		2



How many paths are there for Chris?

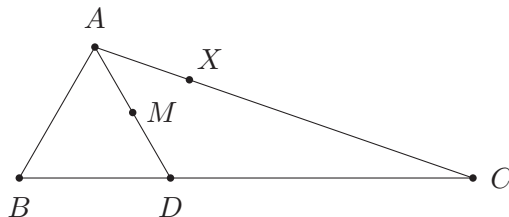
Examination continues overleaf...

Question 12 (14 Marks)

Commence a NEW booklet.

Marks

- (a) The diagram below shows $\triangle ABC$ whose vertices have position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ from an origin O .

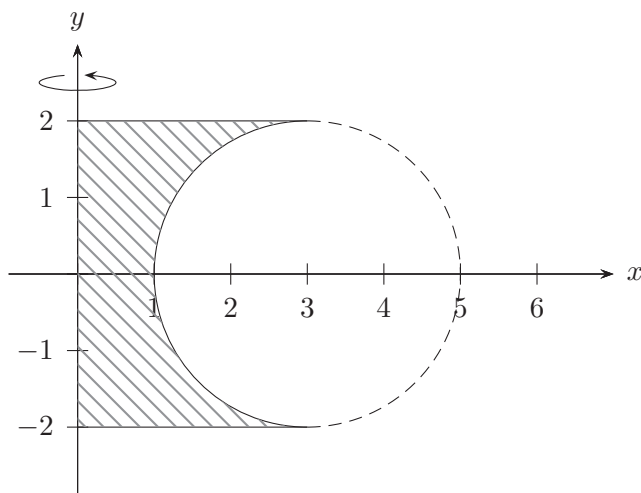


Point D lies on BC such that $BD = \frac{1}{3}BC$, point X lies AC such that $AX = \frac{1}{4}AC$ and M is the midpoint of AD .

- i. Show that $\overrightarrow{AM} = -\frac{1}{2}\underline{a} + \frac{1}{3}\underline{b} + \frac{1}{6}\underline{c}$. **3**
 - ii. Show that B , M and X are collinear. **2**
 - iii. Hence find $BM : MX$. **1**
- (b) i. Show that **1**
- $$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}$$
- ii. Use the substitution $u = \sqrt{x-1}$ to show that **3**

$$\int_2^5 \frac{x-1}{x-1+\sqrt{x-1}} dx = 2 \left(\frac{1}{2} + \ln \frac{3}{2} \right)$$

- (c) To make the rim, the region between the circle $(x-3)^2 + y^2 = 4$, the lines $y = -2$, $y = 2$ and the y -axis is rotated about the y axis. **4**



Find the exact volume of the steel needed to make the rim.

Examination continues overleaf...

Question 13 (13 Marks)

Commence a NEW booklet.

Marks

- (a) On the grid supplied on page 10 and addition of ordinates, sketch the graph of **3**

$$y = x - 2 \sin x \text{ for } 0 \leq x \leq 2\pi$$

Show all important features.

- (b) Find $\int \sin^2 3x \, dx$. **2**

- (c) i. You are given that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. **3**
Use this to show that

$$\sin^4 \theta + \cos^4 \theta = \frac{1}{4}(\cos 4\theta + 3)$$

- ii. By letting $x = \cos \theta$, show that the equation below has roots **3**
 $\pm \cos \frac{\pi}{12}$ and $\pm \cos \frac{5\pi}{12}$.

$$8x^4 + 8(1 - x^2)^2 = 7$$

- iii. Deduce that **2**

$$\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{1}{4}$$

Examination continues overleaf...

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

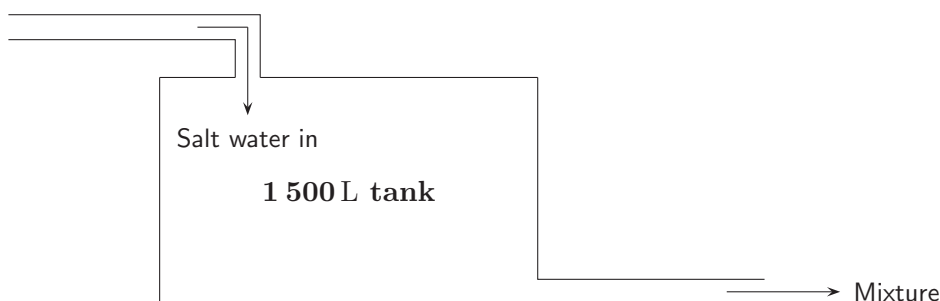
- (a) The velocity of a particle satisfies the differential equation $\frac{dx}{dt} = x \sin(t)$, where x is its displacement in centimetres relative to a fixed point O at time t seconds.

Initially, the displacement of the particle is 1 cm.

- i. Find an expression for x in terms of t . **3**
- ii. Find when the particle first reaches its maximum displacement within $0 \leq t < 2\pi$. **2**

- (b) In the diagram, a tank initially contains 1500 litres of pure water.

Salty water begins pouring into the tank from a pipe, and is stirred to ensure that it is always completely mixed with the pure water.



A second pipe draws the mixture off at the same rate, so that there is always a total of 1500 litres in the tank.

- i. If the salty water entering the tank contains 3 grams of salt per litre, and is flowing in at the constant rate of w litres/min, how much salt is entering the tank per minute? **1**
- ii. If there are Q grams of salt in the tank at time t , how much salt is in 1 litre at time t ? **1**
- iii. Hence write down the amount of salt leaving the tank per minute. **2**
- iv. Use the previous parts to show that $\frac{dQ}{dt} = \frac{w}{1500}(4500 - Q)$. **2**
- v. By solving the differential equation in part (iv), show that **3**

$$Q = 4500 + Ae^{-\frac{wt}{1500}}$$

- vi. Determine the value of A . **1**

End of paper.

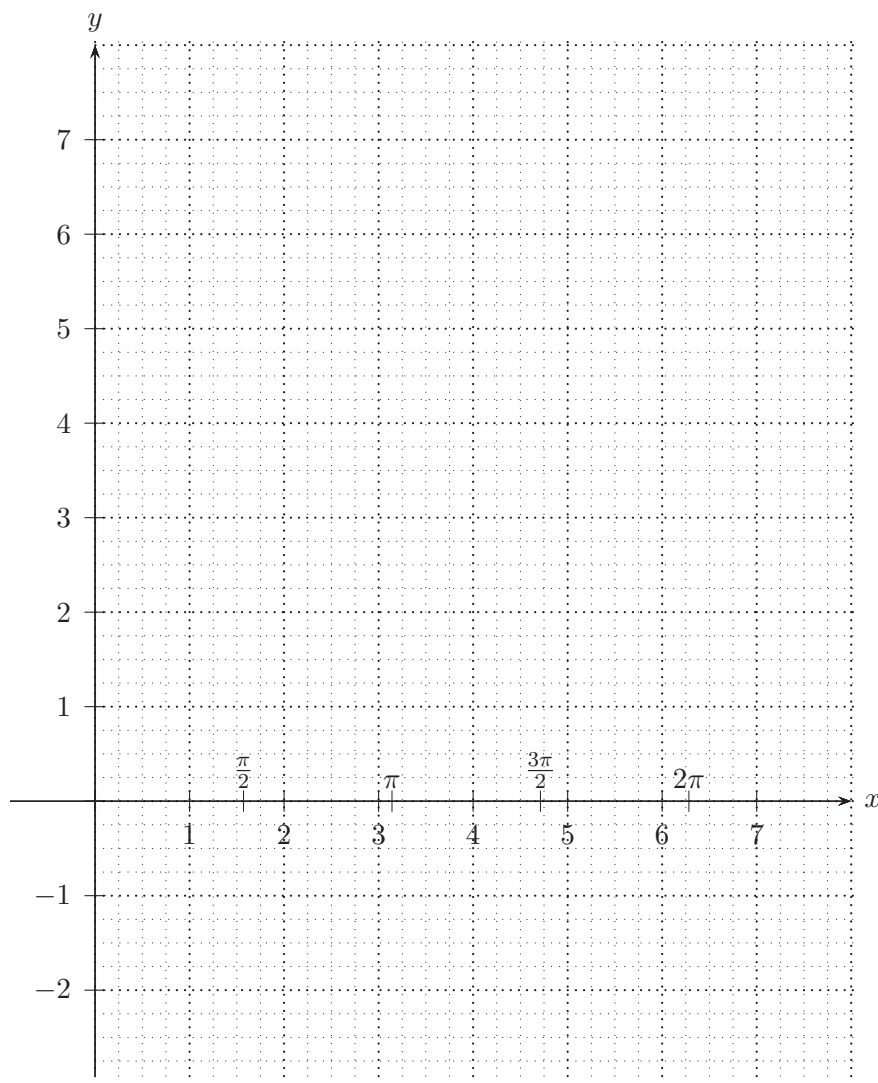
Question 13

- (a) Using the grid supplied below and addition of ordinates, sketch the graph of

3

$$y = x - 2 \sin x \text{ for } 0 \leq x \leq 2\pi$$

Show all important features.



MC

1. B 2. C 3. C 4. A 5. A 6. D 7. B 8. D 9. C 10. B

Short Answer

Question 11. (Miss C. Kim)

$$\begin{aligned} \text{a) i) LHS} &= \frac{1}{(n+1)!} - \frac{n+1}{(n+2)!} \\ &= \frac{n+2}{(n+2)(n+1)!} - \frac{n+1}{(n+2)!} \\ &= \frac{n+2-(n+1)}{(n+2)!} \\ &= \frac{1}{(n+2)!} \\ &= \text{RHS} \end{aligned}$$

ii) Let $P(n)$ be given proposition.

Base case: Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= \frac{1}{(1+1)!} & \text{RHS} &= 1 - \frac{1}{(1+1)!} \\ &= \frac{1}{2} & &= \frac{1}{2} \end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$

$\therefore P(1)$ is true.

Assume true for $n=k$;

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

Prove true for $n=k+1$;

$$\text{RTP: } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

$$\begin{aligned} \text{LHS} &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{By assumption}) \\ &= 1 - \left(\frac{1}{(k+1)!} - \frac{k+1}{(k+2)!} \right) \\ &= 1 - \frac{1}{(k+2)!} \quad [\text{from i)}] \\ &= \text{RHS} \end{aligned}$$

$\therefore P(k+1)$ is true for all $n \in \mathbb{Z}^+$

$\therefore$ True by Mathematical Induction.

$$\text{b) } \left(3x - \frac{1}{x^2} \right)^6 = \sum_{k=0}^6 \binom{6}{k} (3x)^{6-k} (-x^{-2})^k$$

$$\text{Constant term when } x^{6-k} \times x^{-2k} = 1$$

$$6-3k = 0$$

$$3k = 6$$

$$k = 2$$

$$\therefore \text{Constant term} = \binom{6}{2} 3^4 (-1)^2$$

$$= \binom{6}{2} 3^4$$

$$= 1215$$

$$\text{c) i) } {}^{20}C_7 = 77520$$

$$\text{ii) } {}^{18}C_6 = 18564$$

$$\begin{aligned} \therefore P(\text{Jun but not Alex}) &= \frac{{}^{18}C_6}{{}^{20}C_7} \\ &= \frac{91}{380} \end{aligned}$$

d) 5 students, 4 teachers, 2 parents

$$\begin{aligned} \text{i) } & \begin{array}{c} \text{students} \\ \text{P} \quad \text{P} \quad \text{P} \\ \text{P} \quad \text{P} \quad \text{P} \quad \text{P} \quad \text{P} \\ \text{teachers} \end{array} \quad \begin{array}{l} {}^5P_1 \quad {}^4P_1 \quad {}^2P_1 \\ 5! \times 4! \times 2 \\ = 5760 \end{array} \end{aligned}$$

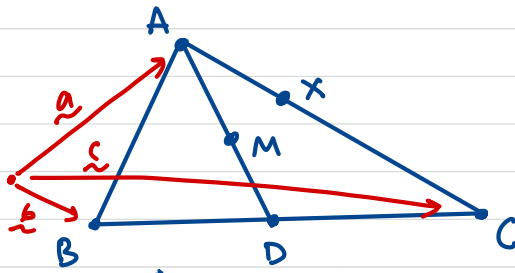
$$\text{ii) } 4! \times 4! \times 2 = 1152$$

$$\begin{aligned} \therefore P(\text{Student sits next to their parent}) &= \frac{1152}{5760} \\ &= \frac{1}{5} \end{aligned}$$

$$\text{e) } \frac{8!}{5!3!} = 56 \quad {}^8C_3 \text{ or } {}^8C_5$$

Question 12 (Mr Lam)

a)



$$i) \vec{AB} = \underline{b} - \underline{a}, \vec{AC} = \underline{c} - \underline{a}, \vec{BC} = \underline{c} - \underline{b}$$

$$\begin{aligned} \vec{BD} &= \frac{1}{3} \vec{BC} \\ &= \frac{1}{3} (\underline{c} - \underline{b}) \end{aligned}$$

$$\begin{aligned} \vec{AD} &= \vec{AB} + \vec{BD} \\ &= \underline{b} - \underline{a} + \frac{1}{3} (\underline{c} - \underline{b}) \\ &= -\underline{a} + \frac{2}{3} \underline{b} + \frac{1}{3} \underline{c} \end{aligned}$$

$$\begin{aligned} \vec{AM} &= \frac{1}{2} \vec{AD} \\ &= \frac{1}{2} \left(-\underline{a} + \frac{2}{3} \underline{b} + \frac{1}{3} \underline{c} \right) \\ &= -\frac{1}{2} \underline{a} + \frac{1}{3} \underline{b} + \frac{1}{6} \underline{c} \end{aligned}$$

$$\begin{aligned} ii) \vec{AX} &= \frac{1}{4} \vec{AC} \\ &= \frac{1}{4} (\underline{c} - \underline{a}) \\ &= \frac{1}{4} \underline{c} - \frac{1}{4} \underline{a} \end{aligned}$$

$$\begin{aligned} \vec{BM} &= \vec{BA} + \vec{AM} \\ &= -\underline{b} + \underline{a} + \left(-\frac{1}{2} \underline{a} + \frac{1}{3} \underline{b} + \frac{1}{6} \underline{c} \right) \\ &= \frac{1}{2} \underline{a} - \frac{2}{3} \underline{b} + \frac{1}{6} \underline{c} \end{aligned}$$

$$\begin{aligned} \vec{MX} &= \vec{MA} + \vec{AX} \\ &= - \left(-\frac{1}{2} \underline{a} + \frac{1}{3} \underline{b} + \frac{1}{6} \underline{c} \right) + \frac{1}{4} \underline{c} - \frac{1}{4} \underline{a} \\ &= \frac{1}{4} \underline{a} - \frac{1}{3} \underline{b} + \frac{1}{12} \underline{c} \\ &= \frac{1}{2} \left(\frac{1}{2} \underline{a} - \frac{2}{3} \underline{b} + \frac{1}{6} \underline{c} \right) \\ &= \frac{1}{2} \cdot \vec{BM} \end{aligned}$$

$$\therefore \vec{MX} \parallel \vec{BM}$$

$\therefore$ B, M and X are collinear.

$$iii) \text{ from ii) } \vec{MX} = \frac{1}{2} \vec{BM}$$

$$\therefore BM : MX = 2 : 1$$

$$b) i) \text{ RHS} = x - 1 + \frac{1}{x+1}$$

$$\begin{aligned} &= \frac{x(x+1) - (x+1) + 1}{x+1} \\ &= \frac{x^2 + x - x - 1 + 1}{x+1} \\ &= \frac{x^2}{x+1} \\ &= \text{LHS} \end{aligned}$$

$$\therefore \frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}$$

$$\begin{aligned} ii) \quad u &= (x-1)^{\frac{1}{2}} & \text{when } x=5, u=2 \\ du &= \frac{1}{2} (x-1)^{-\frac{1}{2}} dx & x=2, u=1 \\ &= \frac{1}{2\sqrt{x-1}} dx \end{aligned}$$

$$\therefore 2u \, du = dx$$

$$\begin{aligned} \int_2^5 \frac{x-1}{x-1+\sqrt{x-1}} dx &= \int_1^2 \frac{u^2}{u^2+u+1} 2u \, du \\ &= 2 \int_1^2 \frac{u^2}{u+1} du \\ &= 2 \int_1^2 \left(u - 1 + \frac{1}{u+1} \right) du \quad (\text{from i}) \\ &= 2 \left[\frac{u^2}{2} - u + \ln(u+1) \right]_1^2 \\ &= 2 \left[\frac{2^2}{2} - 2 + \ln(2+1) - \left(\frac{1^2}{2} - 1 + \ln(1+1) \right) \right] \\ &= 2 \left(\ln 3 + \frac{1}{2} - \ln 2 \right) \\ &= 2 \left(\ln \frac{3}{2} + \frac{1}{2} \right) \end{aligned}$$

$$c) V = \pi \int_a^b x^2 dy$$

$$(x-3)^2 + y^2 = 4$$

$$(x-3)^2 = 4 - y^2$$

$$x-3 = \pm \sqrt{4-y^2}$$

$$x = 3 \pm \sqrt{4-y^2}$$

BUT LHS of semi-circle

$$\therefore x = 3 - \sqrt{4-y^2}$$

$$x^2 = (3 - \sqrt{4-y^2})^2$$

$$= 9 - 6\sqrt{4-y^2} + 4-y^2$$

$$= 13 - 6\sqrt{4-y^2} - y^2$$

$$\therefore V = \pi \int_{-2}^2 13 - 6\sqrt{4-y^2} - y^2 dy$$

$$= 2\pi \int_0^2 13 - 6\sqrt{4-y^2} - y^2 dy$$

$$= 2\pi \int_0^2 13 - y^2 dy - 2\pi \cdot 6 \underbrace{\int_0^2 \sqrt{4-y^2} dy}_{\rightarrow \text{quarter of circle}}$$

$$= 2\pi \left[13y - \frac{y^3}{3} \right]_0^2 - 12\pi \left[\frac{\pi(2)^2}{4} \right]$$

$$= 2\pi \left[26 - \frac{8}{3} \right] - 12\pi^2$$

$$= 2\pi \cdot \frac{70}{3} - 12\pi^2$$

$$= 2\pi \left(\frac{70}{3} - 6\pi \right) \text{ units}^3$$

Question 13 (Miss Lee)

a) check the grid

$$\begin{aligned} b) \int \sin^2 3x \, dx &= \int \frac{1}{2} (1 - \cos 6x) \, dx \\ &= \frac{1}{2} \int 1 - \cos 6x \, dx \\ &= \frac{1}{2} x - \frac{1}{12} \sin 6x + c \end{aligned}$$

c) i) $\cos 4\theta = \cos(3\theta + \theta)$

$$\begin{aligned} &= \cos 3\theta \cos \theta - \sin 3\theta \sin \theta \\ &= (4\cos^3 \theta - 3\cos \theta) \cos \theta - (3\sin \theta - 4\sin^3 \theta) \sin \theta \\ &= 4\cos^4 \theta - 3\cos^2 \theta - 3\sin^2 \theta + 4\sin^4 \theta \\ &= 4(\cos^4 \theta + \sin^4 \theta) - 3(\cos^2 \theta + \sin^2 \theta) \end{aligned}$$

$$\cos 4\theta = 4(\cos^4 \theta + \sin^4 \theta) - 3$$

$$4(\cos^4 \theta + \sin^4 \theta) = \cos 4\theta + 3$$

$$\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{4} (\cos 4\theta + 3)$$

ii) If $x = \cos \theta$,

$$8x^4 + 8(1-x^2)^2 = 7$$

$$8\cos^4 \theta + 8(1-\cos^2 \theta)^2 = 7$$

$$8\cos^4 \theta + 8(\sin^2 \theta)^2 = 7$$

$$8\cos^4 \theta + 8\sin^4 \theta = 7$$

$$\cos^4 \theta + \sin^4 \theta = \frac{7}{8}$$

$$\therefore \frac{1}{4} (\cos 4\theta + 3) = \frac{7}{8}$$

$$\cos 4\theta + 3 = \frac{7}{2}$$

$$\cos 4\theta = \frac{1}{2}$$

$$4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}, \frac{19\pi}{3}, \frac{23\pi}{3}$$

$$\therefore \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$$

$$\cos \frac{\pi}{12} = 0.9659 \dots \quad \cos \frac{5\pi}{12} = 0.2588 \dots$$

$$\begin{aligned} \cos \frac{7\pi}{12} &= \cos \left(\pi - \frac{5\pi}{12} \right) & \cos \frac{11\pi}{12} &= \cos \left(\pi - \frac{\pi}{12} \right) \\ &= -\cos \frac{5\pi}{12} & &= -\cos \frac{\pi}{12} \\ &= -0.2588 \dots & &= -0.9659 \dots \end{aligned}$$

$$\therefore x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, -\cos \frac{5\pi}{12}, -\cos \frac{\pi}{12}$$

$$\text{iii) } 8x^4 + 8(1-x^2)^2 = 7$$

$$8x^4 + 8(1-2x^2+x^4) = 7$$

$$8x^4 + 8x^4 - 16x^2 + 8 - 7 = 0$$

$$16x^4 - 16x^2 + 1 = 0$$

*product of roots;

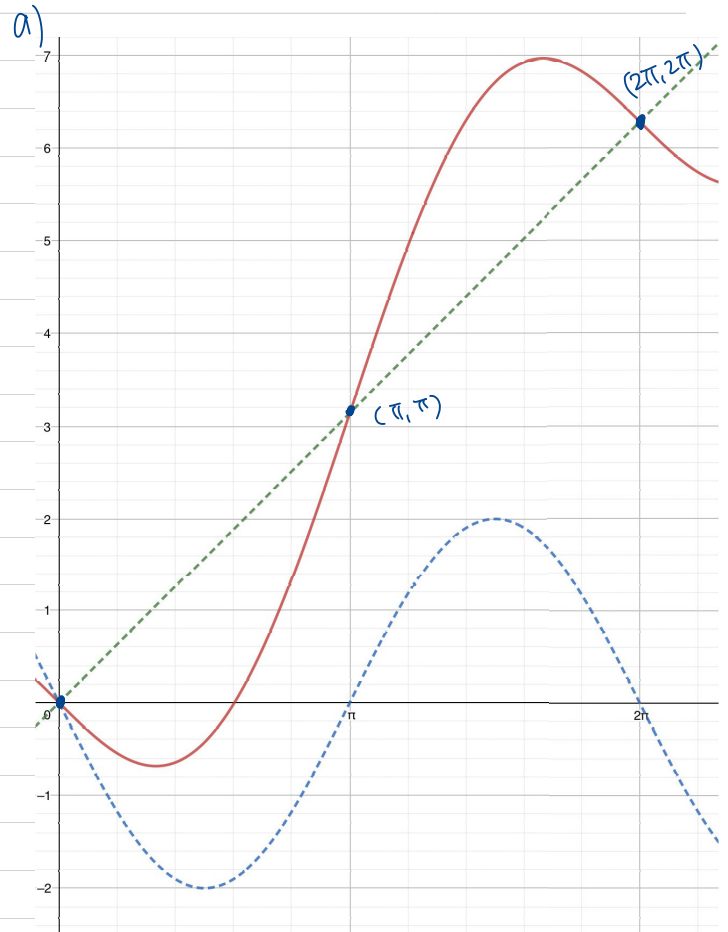
$$\cos \frac{\pi}{12} \times \cos \frac{5\pi}{12} \times -\cos \frac{\pi}{12} \times -\cos \frac{5\pi}{12} = \frac{c}{a} = \frac{1}{16}$$

$$\left(\cos \frac{\pi}{12} \right)^2 \left(\cos \frac{5\pi}{12} \right)^2 = \frac{1}{16}$$

$$\cos \frac{\pi}{12} \cdot \cos \frac{5\pi}{12} = \pm \frac{1}{4}$$

$$\therefore \cos \frac{\pi}{12} > 0 \text{ \& } \cos \frac{5\pi}{12} > 0$$

$$\cos \frac{\pi}{12} \cdot \cos \frac{5\pi}{12} = \frac{1}{4}$$



Question 14 (Mr. Sekaran)

a) i) $\frac{dx}{dt} = x \sin t$

$$\int \frac{1}{x} dx = \int \sin t dt$$

$$\ln |x| = -\cos t + c$$

$$x = e^{-\cos t + c}$$

$$\therefore x = Ae^{-\cos t}$$

when $t=0$, $x=1$;

$$1 = Ae^{-\cos 0}$$

$$1 = Ae^{-1}$$

$$\therefore A = e$$

$$\begin{aligned}\therefore x &= e \cdot e^{-\cos t} \\ &= e^{-\cos t + 1}\end{aligned}$$

ii) Max x when $-\cos t + 1$ is max

$$\therefore t = \pi$$

b) i) $\frac{3g}{L} \times \frac{\omega L}{\text{min}} = 3\omega g/\text{min}$

ii) $\frac{Q}{1500} g/L$

iii) $\frac{Qg}{1500L} \times \frac{\omega L}{\text{min}} = \frac{Q\omega g}{1500 \text{ min}}$

iv) $\frac{dQ}{dt} = \text{in} - \text{out}$

$$= 3\omega - \frac{Q\omega}{1500}$$

$$= \frac{\omega}{1500} (4500 - Q)$$

v) $\frac{dQ}{dt} = \frac{\omega}{1500} (4500 - Q)$

$$\frac{dQ}{dt} = -\frac{\omega}{1500} (Q - 4500)$$

$$\int \frac{1}{Q - 4500} dQ = \int -\frac{\omega}{1500} dt$$

$$\ln |Q - 4500| = -\frac{\omega}{1500} t + c$$

$$Q - 4500 = e^{-\frac{\omega}{1500} t + c}$$

$$Q = 4500 + Ae^{-\frac{\omega t}{1500}}$$

vi) when $t=0$, $Q=0$;

$$0 = 4500 + Ae^0$$

$$\therefore A = -4500$$